Code No: 138AC **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, December - 2020
ADVANCED ALGORITHMS
(Computer Science and Engineering)

Time: 2 hours **Max. Marks: 75**

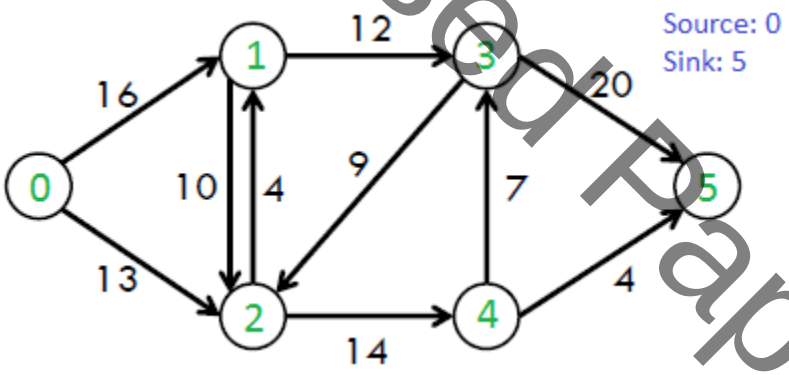
Answer any Five Questions
All Questions Carry Equal Marks
- - -

1.a) Solve the following recurrence equation using substitution method.
$T(n) = 4T(n/2)+n^2$
b) Write the applications of Dynamic Programming. [8+7]

2. Write a Randomized algorithm of Quick Sort and explain. [15]

3. Write an algorithm of Prim's minimum cost spanning tree and illustrate with an example. [15]

4. Find the max flow of the following graph: [15]



5. Explain how to solve the system of linear equations with an example. [15]

6. Explain about the Strassen's Matrix multiplication. [15]

7. Write an algorithm of Rabin-Karp and also analyze the time complexity of the same. [15]

8. Explain about the Approximation algorithm of Vertex cover problem with an example. [15]

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